

Pablo Blasco Fernandez

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Education

PhD Electrical Engineering and Computer Science - MIT

Sept 2026 – Present

- AI for healthcare and medical image analysis, distribution shifts, robustness and clinical deployment in real settings

MSc Biomedical Engineering - ETH Zurich

Sept 2022 – May 2025

- Major in Bioimaging. Relevant coursework: Biomedical Imaging, Neuroinformatics, Image Analysis and Computer Vision, Methods & Models for fMRI Data Analysis - 45 ECTS (50% extra workload) completed in a semester with GPA of 88% (5.5/6)
- Selected as SEMP Scholar twice for research stays at Harvard University

BSc Biomedical Engineering (ranked top 2%) - Carlos III University

Sept 2018 – July 2022

- Semester exchange in Nanyang Technological University of Singapore (KA107 Scholarship - fully covered)
Relevant coursework: Bioimaging, Case studies in bio-pharma-energy, 3D Printing
- Honours in: Electromagnetism, Material Science and Engineering, Linear Algebra, Information Skills, Writing and Communication Skills, Robotics and Bioethics. Students delegate in year 2021/2022
Final BSc Thesis: Association between cerebral perfusion and subclinical cardiovascular disease in middle age: processing and analysis of te-pCASL MRI (Honours)
- Extracurricular courses: Introduction to Artificial Intelligence, Management and leadership & Collaborative teamwork

Experience

Affiliated Researcher - ETH Zurich (Zurich, Switzerland)

June 2025 – Present

- Co-developed Variational Autoencoders and Denoising Diffusion Probabilistic Models (DDPMs) to denoise intra-ventricular monophasic action potential (MAP) signals.
- Designed model architectures and optimized training pipelines to generate robust baselines for electrophysiological data reconstruction.

Affiliated Researcher - Massachusetts Institute of Technology (Boston, MA, USA)

Jan 2025 – Aug 2026

- Engineered deep learning models for Tuberculosis detection from chest X-rays, specializing in out-of-distribution detection and robust classification.
- Partnered with the Sri Lanka Ministry of Health and Colombia Keralty Clinics physicians to curate datasets and embed clinical expertise into model evaluation metrics.
- Streamlined scalable model deployment and reproducible experiment tracking workflows using Hydra and OpenConfig.

AI/ML Data Scientist - Roche (Basel, Switzerland)

June 2025 – July 2026

- Curated data, optimized annotation strategies, and led statistical analysis of mucosal features from Ulcerative Colitis endoscopy videos.
- Developed multi-hierarchical image classification and segmentation models to characterize disease progression.
- Managed large-scale training and validation pipelines using HPC resources, GitLab version control, and experiment tracking.

Graduate Researcher - Harvard University & BWH (Boston, MA, USA)

Sept 2024 – July 2025

- Performed CT segmentation and 3D reconstruction via statistical shape modeling to build patient-specific digital twins, improving simulation precision.
- Developed machine learning classification strategies to predict patient-specific cardiac clinical outcomes.
- Validated model performance and supported clinical decision-making by integrating curated imaging datasets with rigorous statistical analysis.

Data Scientist Intern - Liom AG (Pfaffikon, Switzerland)	March 2024 – Oct 2024
• Extracted actionable insights through comprehensive exploratory data analysis of glucose measurement campaigns. • Boosted key performance indicators by optimizing the predictive accuracy of advanced spectroscopic metrics. • Deployed an outlier detection strategy for live data streams, utilizing statistical distance measures to resolve distributional anomalies.	
Graduate Researcher - Harvard University & MGH (Boston, MA, USA)	Sept 2023 – March 2024
• Developed models and augmentation strategies for automatic cortical/white matter parcellation via 3D-to-2D projections (LEMoN group, Martinos Center).	
Graduate Researcher - ETH Zurich Computer Vision Lab (Zurich, Switzerland)	March 2023 – Aug 2023
• Evaluated the capabilities and performance of training-free neural network architectures for medical imaging applications within the Biomedical Image Computing group.	
Undergraduate Researcher - Spanish National Center for Cardiovascular Research (Madrid, Spain)	June 2021 – July 2022
• Optimized MATLAB and Bash pipelines for MRI/PET brain image analysis (FDG/amyloid) in the PESA population study, collaborating with Dr. Valentín Fuster's research group.	

Publications

Antithetic Sampling Enhanced Denoising Diffusion Probabilistic Models for Complex and Scarce Cardiac Signals	Accepted June 2026
Samuel Ruipérez-Campillo, Pablo Blasco , Moritz Rau, Prasanth Ganesan, Sabyasachi Bandyopadhyay, Albert J. Rogers, Maarten Z.H. Kolk, Fleur V.Y. Tjong, Sanjiv M. Narayan, Julia E. Vogt <i>IEEE Journal of Biomedical and Health Informatics</i>	
Quantifying the Spatial Inter-Rater Reliability of Mucosal Lesion Annotations in Ulcerative Colitis	May 2026
Pablo Blasco Fernandez , Peter Bossuyt, Marco Daperno, Uri Kopylov, Yoram Bouhnik, Konstantinos Karmiris, Fethallah Benmansour, Benjamin Gutierrez Becker, Stefan Fraessle, Bernhard Stimpel, Steven Levitte, Alvaro Gomariz <i>Gastrointestinal Endoscopy 103 (5), S-550-S-551 (DDW 2026)</i>	
Quantifying Inter-Rater Reliability of Mucosal Features in Ulcerative Colitis Endoscopy	Jan 2026
Pablo Blasco Fernandez , Peter Bossuyt, Marco Daperno, Uri Kopylov, Yoram Bouhnik, Konstantinos Karmiris, Fethallah Benmansour, Benjamin Gutierrez Becker, Stefan Fraessle, Bernhard Stimpel, Steven Levitte, Alvaro Gomariz <i>European Crohn's and Colitis Organisation (ECCO) 2026 Conference - Oral Spotlight</i>	
Pseudo-rendering for Resolution and Topology-Invariant Cortical Parcellation	Oct 2024
Pablo Blasco Fernandez , Karthik Gopinath, John Williams-Ramirez, Rogeny Herisse, Lucas Deden-Binder, Dina Zemlyanker, Theresa Connors, Liana Kozanno, Derek Oakley, Bradley Hyman, Sean Young, Juan Iglesias <i>International Workshop on Machine Learning in Medical Imaging (MLMI) 2024 - Endorsed by MICCAI 2024 Conference</i>	

Institutional Responsibilities and Service

AI4MED - Corporate Relations Team Lead	Nov 2023 - Feb 2024
BEST UC3M - Corporate Relations Department Staff Member	Sept 2020 - Sept 2022
TEDxUC3M - Speakers Coordinator (Board of Directors)	Sept 2020 - Sept 2022
Isabel La Catolica School - Volunteering Teaching Assistant	Sept 2015 - May 2017

Awards and Honors

Spotlight Speaker - Royal Complutense College at Harvard University	2025
Awarded - Nova 111 List (Spain) - Nationwide Ranking Top 10 Engineering Students in Engineering	2024
ETH Zurich SEMP Scholarship - Stipend for research studies abroad at Harvard University	2023,2024
Awarded - 'la Caixa' Fellowship (Spain & Portugal) - Full tuition and stipend for any university worldwide, 11/100 awardees within 'Engineering and Technology'	2022
KA107+ Scholarship - Non-European full-tuition mobility scholarship for exchange at NTU Singapore	2021
Spanish Ministry of Education Merit Award - Prize for excellent academic records	2021
Gold Medal - EIT Health Innovation-Days (Spain) - Best health startup proposal	2021
Bronze Medal - EIT Health Innovation-Days (Europe) - Best affordable health solution	2021
Bronze Medal - Engineering Medicine Contest (Madrid) - Top 3 case study solutions	2021
Romanillos Foundation Excellence Scholarship - Full tuition fees for low-income students	2018 to 2021

Misc

Selected Attendee - MIT Summer Geometry Initiative Tutorial Week	2025
Selected Attendee - UCL Medical Image Computing Summer School	2025
Selected Attendee - RISE-MICCAI Medical Imaging Summer School	2025

Skills and Languages

Technologies: Python (NumPy, Pandas, Matplotlib, Seaborn, OpenCV, scikit-learn, PyTorch, Lightning, Weights & Biases), R (RStudio), MATLAB, Blender, git, High-performance Computing

Languages: Spanish (Native), English (Native), German (A2), French (A1)